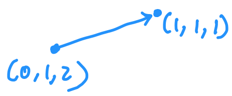
Question 1. Find a vector $\vec{v} \in \mathbb{R}^3$ that satisfies each of the following conditions.

(1) $\vec{v}$ points from $(0,1,2)$ to $(1,1,1)$.

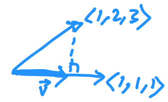
$(1,1,1)-(0,1,2)=(1,0,-1)$

$\vec{v}=\langle 1,0,-1\rangle$



(2) $\vec{v}$ is the projection of $\langle 1,2,3\rangle$ onto $\langle 1,1,1\rangle$.

Normalize $\langle 1,1,1\rangle \longrightarrow \frac{\langle 1,1,1\rangle}{\|\langle 1,1,1\rangle\|}=\frac{1}{\sqrt{3}}\langle 1,1,1\rangle =: \hat{n}$

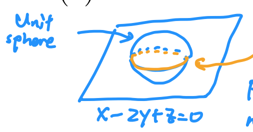


$\langle 1,2,3\rangle\cdot\hat{n}=\langle 1,2,3\rangle\cdot\frac{1}{\sqrt{3}}\langle 1,1,1\rangle=\frac{6}{\sqrt{3}}$

$\vec{v}=(\langle 1,2,3\rangle\cdot\hat{n})\,\hat{n}=\frac{6}{\sqrt{3}}\cdot\frac{1}{\sqrt{3}}\langle 1,1,1\rangle=\langle 2,2,2\rangle$

Projection formula: $(\vec{v}\cdot\hat{n})\hat{n}$ for unit vector $\hat{n}$; $(\vec{v}\cdot\frac{\vec{u}}{\|\vec{u}\|})\frac{\vec{u}}{\|\vec{u}\|}$ for any vector $\vec{u}$

(3) $\vec{v}$ is a unit vector in the plane $x-2y+z=0$.



A whole circle worth of answers!

Find any nonzero vector in the plane and normalize.

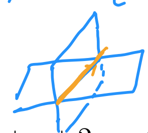
For example, $\langle 1,0,-1\rangle$ is in the plane, and we get $\vec{v}=\frac{1}{\sqrt{2}}\langle 1,0,-1\rangle$ after normalization.

(4) $\vec{v}\neq 0$ is in the intersection of the plane $x-2y+z=0$ and the plane $2x+y-z=0$.

Set $\vec{v}=\langle x,y,z\rangle$, then we just need to find nontrivial solution to $\begin{cases}x-2y+z=0\\2x+y-z=0\end{cases} \Longleftrightarrow \begin{cases}x-2y+z=0\\5y-3z=0\end{cases} \Longleftrightarrow \begin{cases}x-\frac{1}{5}z=0\\y-\frac{3}{5}z=0\end{cases}$

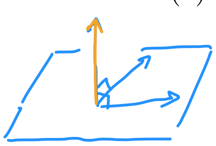
If we set $z=5$, then $x=1$, $y=3$.

So $\vec{v}=\langle 1,3,5\rangle$ is a solution.



A line worth of answers!

(5) $\vec{v}\neq 0$ is a vector orthogonal to $\langle 1,-2,1\rangle$ and to $\langle 2,1,-1\rangle$.



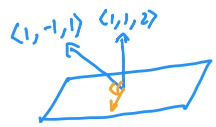
$\vec{v}=\langle 1,-2,1\rangle\times\langle 2,1,-1\rangle=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-2&1\\2&1&-1\end{vmatrix}$

$=\begin{vmatrix}-2&1\\1&-1\end{vmatrix}\hat{i}-\begin{vmatrix}1&1\\2&-1\end{vmatrix}\hat{j}+\begin{vmatrix}1&-2\\2&1\end{vmatrix}\hat{k}=\langle 1,3,5\rangle$

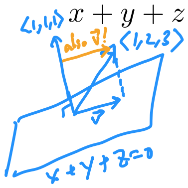
Same answer as (4)!
Being orthogonal to $\langle a,b,c\rangle \Longleftrightarrow$ in the plane $ax+by+cz=0$

(6) $\vec{v}\neq 0$ is a vector in the plane $x+y+2z=0$ and is orthogonal to $\langle 1,-1,1\rangle$.

$\vec{v}=\langle 1,1,2\rangle\times\langle 1,-1,1\rangle=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&2\\1&-1&1\end{vmatrix}=\langle 3,1,-2\rangle$



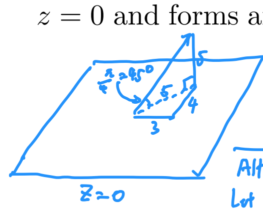
(7) $\vec{v}$ is the projection of $\langle 1,2,3\rangle$ onto the plane $x+y+z=0$.



$\vec{v}$ points from $\mathrm{proj}_{\langle 1,1,1\rangle}\langle 1,2,3\rangle=\langle 2,2,2\rangle$ (part (2)) to $\langle 1,2,3\rangle$.

So $\vec{v}=\langle 1,2,3\rangle-\langle 2,2,2\rangle=\langle -1,0,1\rangle$

(8) (challenge) $\vec{v}$ is a unit vector in the plane $z=0$ and forms an angle of $\frac{\pi}{4}$ with $\langle 3,4,5\rangle$.



We can see that $\langle 3,4,0\rangle$ is pointing in the right direction, and we get $\vec{v}=\frac{1}{5}\langle 3,4,0\rangle$ after normalization.

Alternatively, we can just do algebra.
Let $\vec{v}=\langle x,y,0\rangle$ with $\|\vec{v}\|=x^2+y^2=1$. ($z=0$)

Law of cosine: $\frac{1}{\sqrt{2}}=\cos\frac{\pi}{4}=\frac{\vec{v}\cdot\langle 3,4,5\rangle}{\|\vec{v}\|\cdot\|\langle 3,4,5\rangle\|}=\frac{3x+4y}{\sqrt{50}}$

$\Rightarrow 3x+4y=5 \Rightarrow y=\frac{1}{4}(5-3x)$

$x^2+y^2=1 \Rightarrow x^2+\frac{1}{16}(5-3x)^2=1$

$\Rightarrow 16x^2+(5-3x)^2=16$

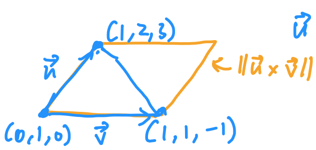
$\Rightarrow 25x^2-30x+9=0$

$\Rightarrow (5x-3)^2=0$

$\Rightarrow x=3/5 \quad y=4/5$ phew! ö

Question 2. Find the area or volume of the following shapes in $\mathbb{R}^3$.

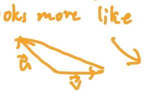
(1) The area of the triangle with vertices $(0,1,0)$, $(1,1,-1)$, and $(1,2,3)$.



$\vec{u}=\langle 1,1,3\rangle \qquad \vec{v}=\langle 1,0,-1\rangle$

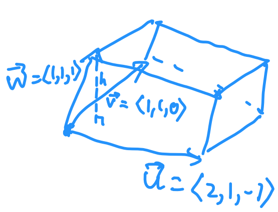
Area $=\frac{\|\vec{u}\times\vec{v}\|}{2}=\frac{\|\langle -1,4,-1\rangle\|}{2}=\frac{3\sqrt{2}}{2}$

In fact, it looks more like



Really doesn't matter how I place the vertices, it's just visual aid.

(2) The volume of the parallelepiped with a vertex at the origin with adjacent vertices at $(2,1,-1)$, $(1,1,0)$, $(1,1,1)$.



Volume $=\vec{w}\cdot(\vec{u}\times\vec{v})=\vec{w}\cdot\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&-1\\1&1&0\end{vmatrix}=\begin{vmatrix}1&1&1\\2&1&-1\\1&1&0\end{vmatrix}=\begin{vmatrix}1&-1\\1&0\end{vmatrix}-\begin{vmatrix}2&-1\\1&0\end{vmatrix}+\begin{vmatrix}2&1\\1&1\end{vmatrix}$

$=1-1+1$

$=1$